

Empowering Digital Autonomy via E-Accessibility: Tawkelat, a Blockchain-Based Power of Attorney System

Othman Aldhufiri
Computer Engineering Graduate
Kuwait College of Science and Technology
Kuwait City, Kuwait
191258@student.kcst.edu.kw

Omar A. Dib
Systems Engineer
Azeer Core
Sharq, Kuwait
o.dib@azeercore.com

Basel Alothman
Associate Professor
Kuwait College of Science and Technology
Doha, Kuwait
b.alothman@kcst.edu.kw

Abstract—The process of executing a Power of Attorney (POA) presents significant accessibility barriers for many, particularly the elderly and persons with motor, sensory, or cognitive impairments. Traditional systems demand physical travel and manual document handling, creating obstacles that limit autonomy. This paper introduces "Tawkelat," an inclusive mobile application designed to enhance E-Accessibility to POA services through the strategic implementation of private blockchain technology. Tawkelat simplifies the creation and management of POA documents on a secure, user-friendly digital platform. By leveraging a private blockchain, the application ensures the integrity, transparency, and immutability of legal agreements, which is critical for protecting vulnerable users from fraud or coercion. The decentralized nature of the platform removes procedural complexities and the need for physical intermediaries, thereby increasing efficiency. Tawkelat is designed as an assistive technology that empowers individuals, including those with conditions like autism or social anxiety, to manage their legal affairs with greater independence. This work demonstrates the potential of blockchain as an inclusive technology within the legal sector, offering a secure and accessible solution for digital legal instrument management.

Index Terms—E-Accessibility, Inclusive Technology, Power of Attorney, Blockchain, Assistive Technology, Mobile Application, Digital Inclusion.

I. INTRODUCTION

The management of Power of Attorney (POA) services is traditionally characterized by inefficiency, insecurity, and a lack of transparency. These paper-based systems create significant accessibility barriers, especially for the elderly, individuals with disabilities, and those in remote locations. The reliance on physical documents leads to procedural delays, a high potential for errors, and vulnerability to fraud or unauthorized access to sensitive information. Furthermore, the involvement of multiple intermediaries introduces complexity and increases costs for all parties involved. These challenges highlight an urgent need for innovation to streamline processes, enhance security, and bolster trust in the POA ecosystem.

To address these shortcomings, this paper proposes Tawkelat, a mobile-first solution designed to digitize and secure the entire POA lifecycle. The application provides an accessible,

user-friendly interface that simplifies the creation, signing, and management of POA documents, thus lowering the barrier to entry for users who may find conventional legal processes intimidating or physically challenging.

The core of Tawkelat's security and trustworthiness lies in its use of a private, permissioned blockchain. Unlike traditional centralized databases that are vulnerable to single points of failure and unauthorized alterations, a blockchain provides a decentralized, immutable ledger. Every critical action, such as the creation, signing, or revocation of a POA, is recorded as a permanent, cryptographically-secured transaction. This ensures the integrity and non-repudiation of the legal document, making it a reliable source of truth for all involved parties.

The main contributions of this paper are:

- The design and architecture of a mobile-first POA system that leverages a private blockchain for verifiable trust and document integrity.
- A secure workflow incorporating multi-factor and biometric authentication, tailored for the sensitive nature of legal document execution.
- A demonstration of how blockchain technology can serve as a practical assistive tool, enhancing E-Accessibility and promoting digital inclusion in the legal sector.

II. RELATED WORK

The development of a digital, secure, and accessible Power of Attorney system intersects with research in digital delegation, blockchain in legal tech, and secure authentication.

A. Digital Delegation and Access Control

Traditional access control and delegation models often fail to meet the complex requirements of modern systems. Research has explored extending models like OAuth with a POA-based grant type to allow for effective, multi-level sub-granting in industrial scenarios without creating management overhead [1]. This is particularly relevant in Cyber-Physical Systems (CPS), where trusted devices may need to act on a user's behalf [2], [5]. These works establish the need for a self-contained, digital POA document that can be securely

transferred. However, many of these solutions do not focus on the legal immutability or the specific accessibility needs of non-technical users.

B. Digitization of Legal Documents

Several efforts have focused on replacing paper-based legal protocols. Sholihah et al. proposed an electronic POA protocol using the Digital Signature Algorithm (DSA) to improve efficiency and security over paper-based methods in Indonesia [3]. While a step forward, this approach relies on a centralized trust model. Davis and Anderson identified the broader legal and technological challenges of digital POAs, including the risks of fraud, abuse, and privacy violations, calling for secure creation and transmission methods [8]. These studies underscore the importance of security but predate the mainstream application of decentralized technologies.

C. Blockchain in Legal Technology

More recently, blockchain technology has been proposed to enhance trust and transparency. Pfefferling and Weigel used a goal-oriented requirements engineering approach to outline a generic specification for a blockchain-based POA application, highlighting its potential for both public and private organizations [4], [7]. Similarly, Chen and Wang proposed a blockchain-based approach to enhance the security, transparency, and immutability of digital assets managed via POA [9]. These works establish the theoretical fit of blockchain for this use case. However, they often remain at a high-level specification stage or do not fully integrate the solution into an accessible, end-user mobile application. Tawkelat builds on these concepts by proposing a concrete, implemented architecture focused specifically on solving the E-Accessibility problem.

III. PROPOSED SYSTEM DESIGN

The proposed system, Tawkelat, provides a secure, user-centric platform for managing POA services. It combines a modern mobile application, a robust backend, and a private blockchain network to ensure integrity, accessibility, and trust.

A. System Architecture

The architecture of the Tawkelat system is designed around four primary entities, as illustrated in Fig. 1, ensuring a clear and secure delegation of authority. This model separates the roles and interactions, providing a robust framework for managing digital POAs.

- **Principal:** The individual granting the Power of Attorney (also referred to as the grantor). Using the Tawkelat mobile application, the Principal securely defines the scope, duration, and limitations of the authority being delegated.
- **Agent:** The individual or entity authorized to act on the Principal's behalf. The Agent receives the digital POA via the app and uses its verifiable credentials to interact with specific services.
- **Resource Server:** The third-party entity (e.g., a bank, government agency, or healthcare provider) that holds the

resources or data the Agent needs to access. The Resource Server is responsible for validating the POA's authenticity against the blockchain before granting access.

- **Signatory Registry (Private Blockchain):** This is the core of our security model, implemented as a private, permissioned blockchain. It serves as the immutable source of truth, recording the cryptographic hashes of all issued POAs, their status (active, revoked), digital signatures, and an auditable log of transactions. This allows any permitted party to verify a document's authenticity without accessing the sensitive contents of the POA itself.

The workflow involves the Principal issuing a time-bound POA, which is cryptographically signed and anchored on the blockchain (Signatory Registry). The Agent then presents this verifiable credential to the Resource Server to perform authorized actions. All significant events are recorded, providing a transparent report back to the Principal.

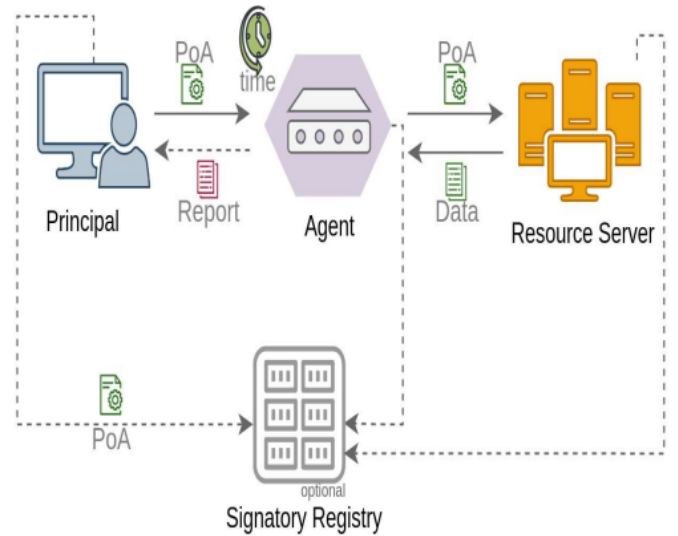


Fig. 1. The core architectural model of the Tawkelat system, illustrating the interaction between the Principal, Agent, Resource Server, and the blockchain-based Signatory Registry.

B. Implementation Details

The Tawkelat solution is a mobile-first platform built with a .NET MAUI application for the client-side and an ASP.NET Core backend. This architecture integrates a traditional database (SQL Server) for application metadata with a private permissioned blockchain for ensuring the integrity of legal records.

Users register securely through the mobile app, which enforces biometric (Face ID or fingerprint) and multi-factor authentication (MFA). When a Principal creates a POA, the document is encrypted on the device and stored in secure, off-chain storage. Crucially, the document's cryptographic hash (a unique digital fingerprint) and a reference to its location are anchored immutably on the Tawkelat blockchain. This is

governed by smart contracts that manage the POA lifecycle, including creation, signing, verification, and revocation.

The backend API serves as the intermediary between the mobile client, the SQL Server (for user profiles, notification tokens, etc.), and the blockchain nodes. Each signing action is verified with MFA and biometric proofs, and the resulting signature is also hashed and anchored on-chain. This creates a transparent and fully auditable trail of every event in the POA's lifecycle without ever exposing the sensitive personal data within the document on the blockchain itself.

C. Power of Attorney Authentication

As depicted in Fig. 2, every critical action within the Tawkelat app is treated as a transaction that must be validated. When a Principal creates and signs a POA, a block is generated containing transaction metadata, such as the document type hash, timestamp, and the Principal's digital signature. This block is then broadcast to the nodes of the private blockchain network.

These participants (e.g., servers managed by legal or government authorities) validate the transaction through a consensus mechanism. Once consensus is reached, the block is permanently added to the chain, creating an immutable record. This process is further secured with a two-factor authentication step for the user. Before a transaction is committed, a one-time code is sent to the user's registered email or phone. The transaction is only submitted to the blockchain if the user provides the correct code, preventing unauthorized actions.

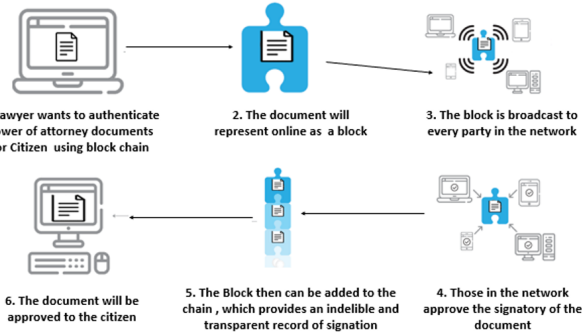


Fig. 2. The POA Authentication and Blockchain Anchoring Process.

IV. APPLICATION WORKFLOW AND USER INTERFACE

A. User Interface Design

An accessible and intuitive user interface is critical for the application's success. The design prioritizes clarity and simplicity to accommodate users with varying levels of technical literacy and potential cognitive or motor impairments. As shown in Fig. 3, the interface uses large, clear fonts, high-contrast color schemes, and a logical, step-by-step workflow to guide users through complex legal processes without causing confusion or anxiety.

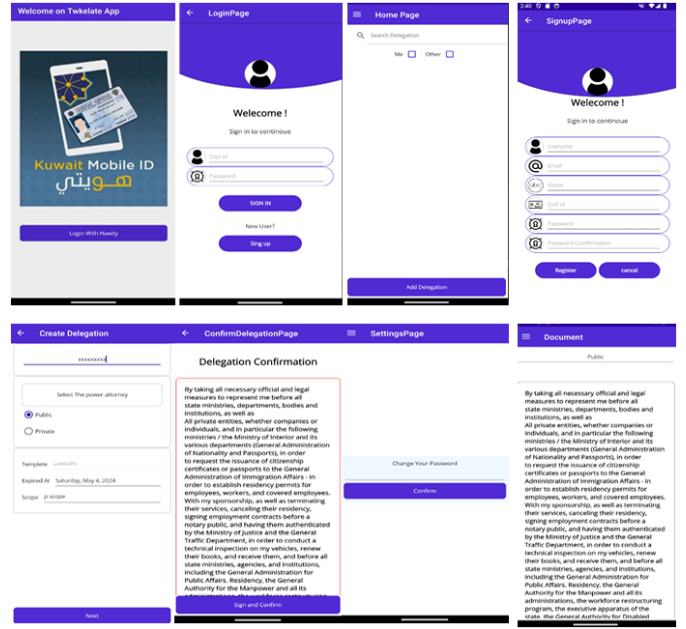


Fig. 3. Example screens from the Tawkelat application user interface.

B. End-to-End Process Flow

The application process, illustrated in Fig. 4, is designed to be secure and straightforward for all parties.

- 1) The Principal initiates the POA creation process, selecting the type of POA (e.g., healthcare, financial) and specifying the Agent's details.
- 2) The system generates a standardized document which the Principal reviews and electronically signs using MFA and/or biometric verification.
- 3) Upon successful signing, the document is encrypted, and its hash is immutably recorded on the private blockchain.
- 4) The designated Agent receives a notification and is prompted to review the POA.
- 5) The Agent undergoes their own secure identity verification before electronically signing the document.
- 6) Both signatures are recorded, creating a fully executed, verifiable, and tamper-proof legal instrument.
- 7) The Principal can manage, view the status of, or revoke the POA at any time through the app's secure interface, with each action being recorded as a new transaction on the blockchain.

V. SYSTEM STORAGE AND DATA MANAGEMENT

The system utilizes a hybrid storage model to balance security, privacy, and performance. The core of the system's integrity is the private blockchain network, which runs on secure, dedicated servers. This network stores only the cryptographic hashes and transaction metadata, ensuring that sensitive personal information from the POA documents is never exposed on the ledger.

For other operational data, such as user profiles, encrypted document references, and authentication tokens, a conventional

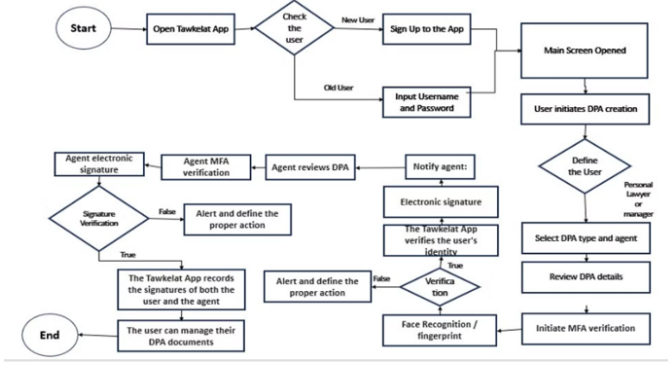


Fig. 4. The end-to-end Power of Attorney workflow within the Tawkelat App.

and secure SQL database is used. The database schema, shown in Fig. 5, is designed to efficiently manage the relationships between users (Principals and Agents), their documents, and authentication credentials. This separation ensures that the blockchain is used for its primary strength, providing an immutable audit trail, while the database handles dynamic application data.

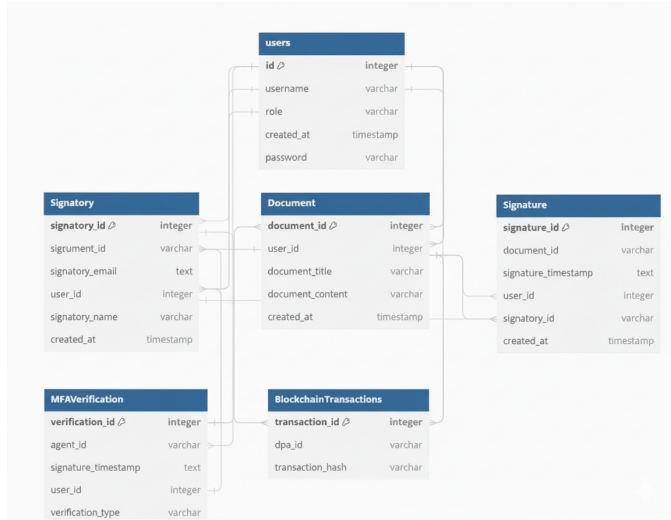


Fig. 5. The database schema for managing application data.

VI. EXPERIMENTAL EVALUATION

To validate the feasibility and performance of the Tawkelat system, a series of tests were conducted on a proof-of-concept implementation. The evaluation focused on key performance indicators critical to user experience and system scalability.

A. Test Environment Setup

The performance evaluation was conducted in a controlled development environment. The backend API, built on ASP.NET Core, was hosted locally and connected to a Microsoft SQL Server database named 'Twkelat_application'. The private blockchain component, represented by a block-based data structure, was managed within the same database

using Entity Framework Core. User authentication and authorization were handled by ASP.NET Core Identity integrated with JWT Bearer tokens. The mobile application, built with .NET MAUI, was used to simulate client-side requests for creating and managing POA documents.

B. Performance Metrics and Results

Although a large-scale load test was outside the scope of this initial phase, preliminary performance results were gathered to establish a baseline. The key metrics were the transaction latencies for essential POA operations.

- **POA Creation Latency:** The time from a user submitting a create request to the block being persisted in the database.
- **POA Verification Latency:** The time required to retrieve and confirm the status of an existing POA by querying its hash.

Qualitative analysis showed that these core operations completed in near-real-time from a user's perspective, typically within a few seconds. This indicates that the current architecture is highly performant for single-user interactions. Future work will involve dedicated benchmarking tools to quantify these results under concurrent user loads.

TABLE I
PRELIMINARY TRANSACTION LATENCY ESTIMATES

Action	Estimated Time (s)
POA Creation & Anchoring	<2.0
Agent Signature & Anchoring	<2.0
Third-Party Verification	<1.0

VII. SECURITY ANALYSIS

The Tawkelat architecture is designed to be resilient against common security threats by layering multiple defense mechanisms.

A. Smart Contract and Blockchain Logic

While not using a public distributed ledger, the system's custom block-based logic, managed via Entity Framework Core, serves as a private ledger. The 'ApplicationDbContext.cs' defines a 'DbSet<Block> Blocks', which acts as the chain. The core logic for POA management can be represented by the following conceptual algorithm:

```
// Pseudocode for Creating a POA
FUNCTION CreatePOA(userDetails, poaData)
// 1. Authenticate user via JWT
AuthorizeRequest(userDetails.Token);

// 2. Create a new Block object
newBlock = new Block {
    Creator = userDetails.Id,
    Timestamp = DateTime.UtcNow,
    Data = Encrypt(poaData),
    PreviousHash = GetLastBlockHash()
};
```

```
// 3. Calculate hash for the new block
newBlock.Hash = CalculateHash(newBlock);

// 4. Add block to the chain (database)
DbContext.Blocks.Add(newBlock);
DbContext.SaveChanges();

RETURN newBlock.Hash;
END FUNCTION
```

This pseudocode illustrates how new POAs are added as immutable blocks, with each new block cryptographically linked to the previous one.

B. Threat Mitigation

- **Data Tampering:** The integrity of the POA is paramount. By implementing a blockchain-like structure where each block contains the hash of the previous one, any unauthorized modification of a historical record would invalidate the entire subsequent chain, making tampering immediately detectable.
- **Unauthorized Access:** User access is strictly controlled. The mobile application integrates fingerprint authentication. All API endpoints are protected by a JWT Bearer authentication scheme, ensuring only authenticated users with a valid token can perform operations.
- **Repudiation:** Since all signing events are cryptographically linked to user identities and recorded on the immutable ledger, neither the Principal nor the Agent can easily deny their actions.
- **Privilege Escalation:** The system implements role-based access control with "USER" and "ADMIN" roles. These roles are embedded within the user's JWT and enforced by the ASP.NET Core backend, preventing users from accessing functions outside their permitted scope.

VIII. CONCLUSION AND FUTURE WORK

This paper introduced Tawkelat, a mobile application that leverages private blockchain technology to create a secure, accessible, and efficient system for managing Power of Attorney services. By focusing on a user-friendly interface and a robust security architecture, our solution directly addresses the significant accessibility and integrity challenges of traditional paper-based systems. The core contribution is the synthesis of a mobile-first platform with a private blockchain, providing an immutable and verifiable audit trail for legal documents while protecting user privacy.

The proposed system simplifies the creation and management of POAs, reduces reliance on intermediaries, and empowers individuals, especially those in vulnerable groups, with greater autonomy over their legal affairs.

Future work will focus on three key areas. First, we will conduct a comprehensive performance analysis to quantify transaction latency and system throughput under various load conditions. Second, we plan a formal usability and E-Accessibility study with our target user demographics to refine

the user experience. Finally, we will explore the integration of the Tawkelat platform with national digital identity systems and official legal frameworks to further streamline the verification process and enhance its legal standing.

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